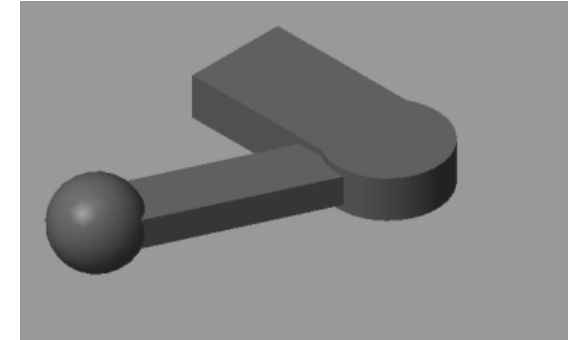
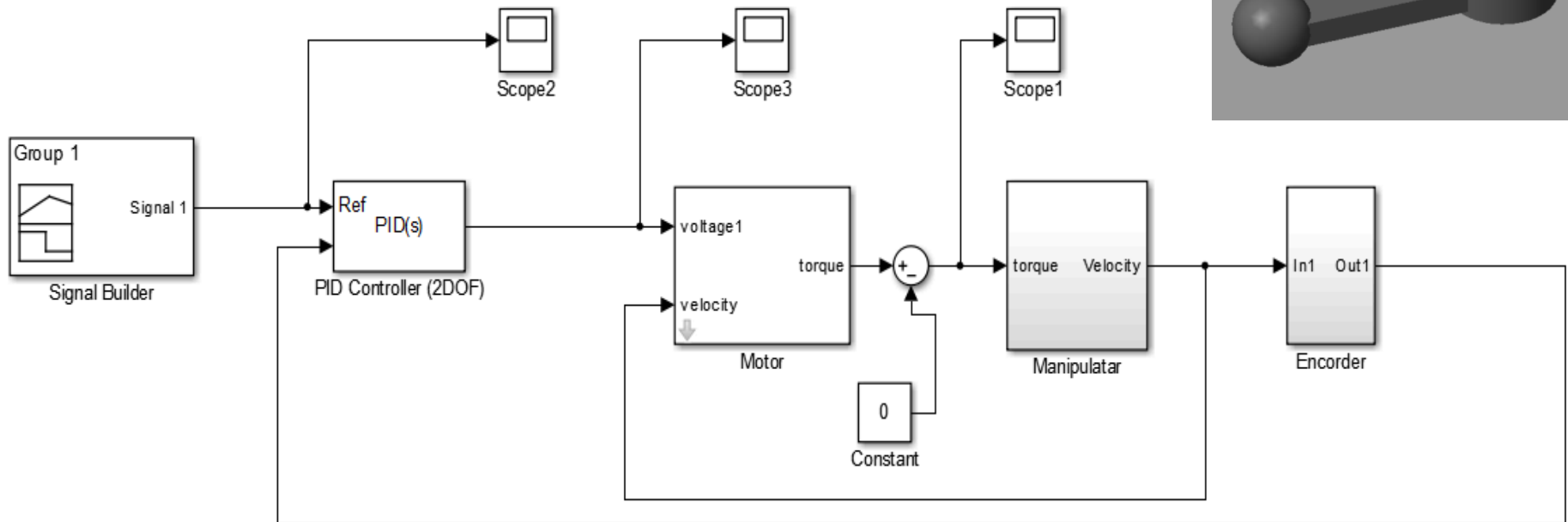


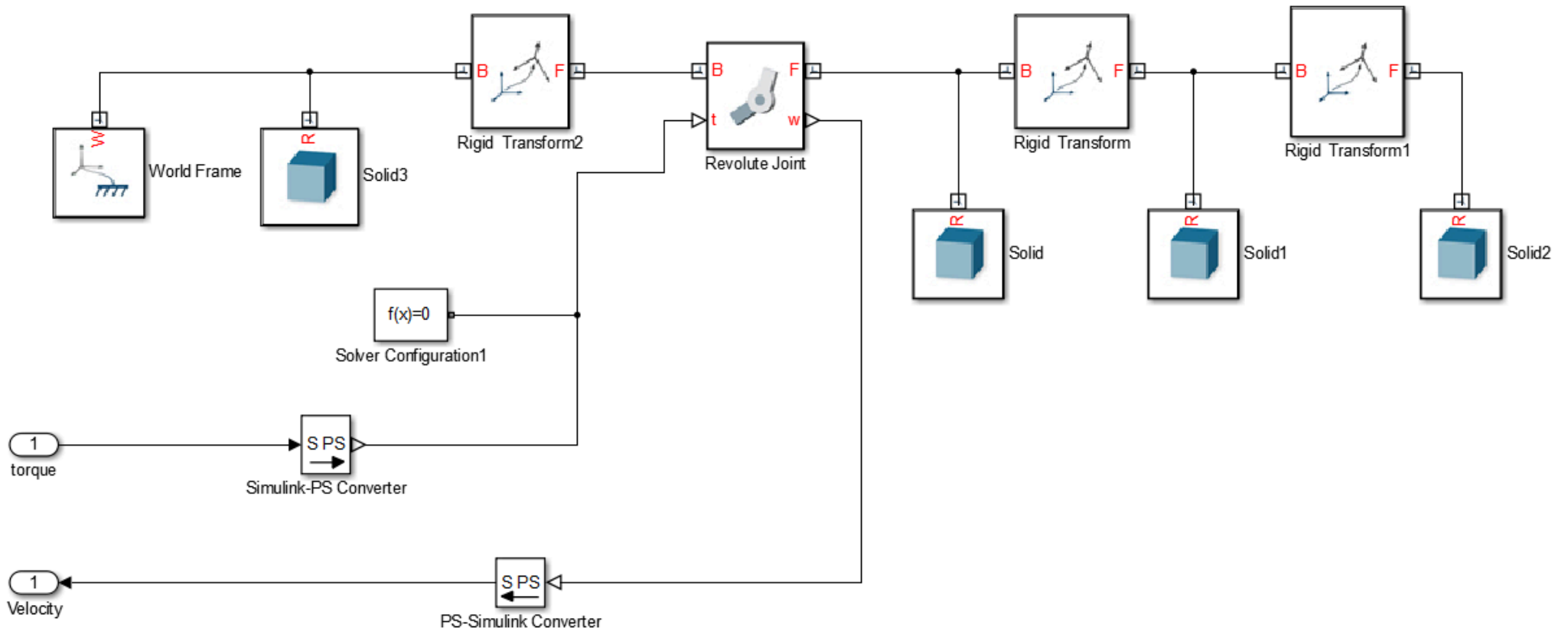
Example 1: 1 D robot manipulator with a controller

1. Overall system:



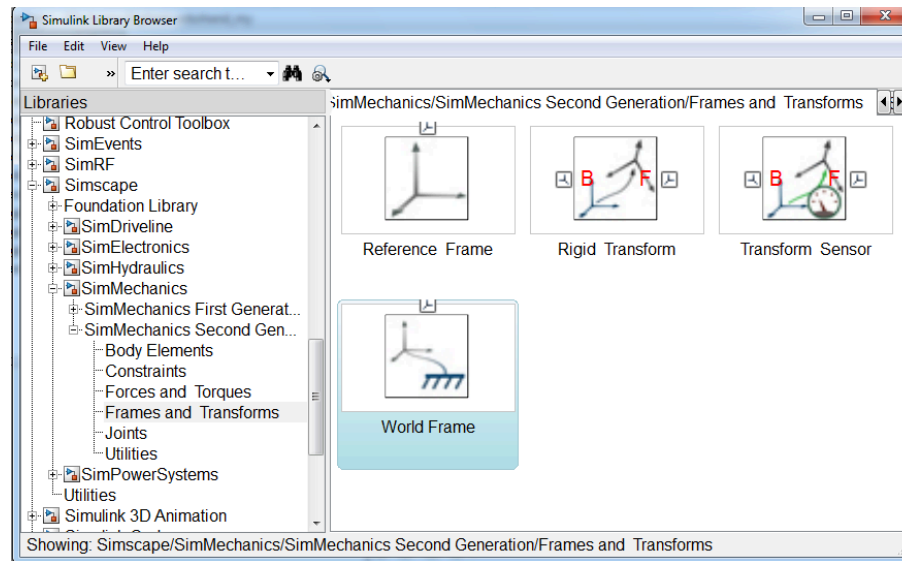
The overall system is developed by integrating the models of the controller, mechanical structure, electrical actuator (motor). You can these sections and combine to develop the overall system.

2. Manipulator



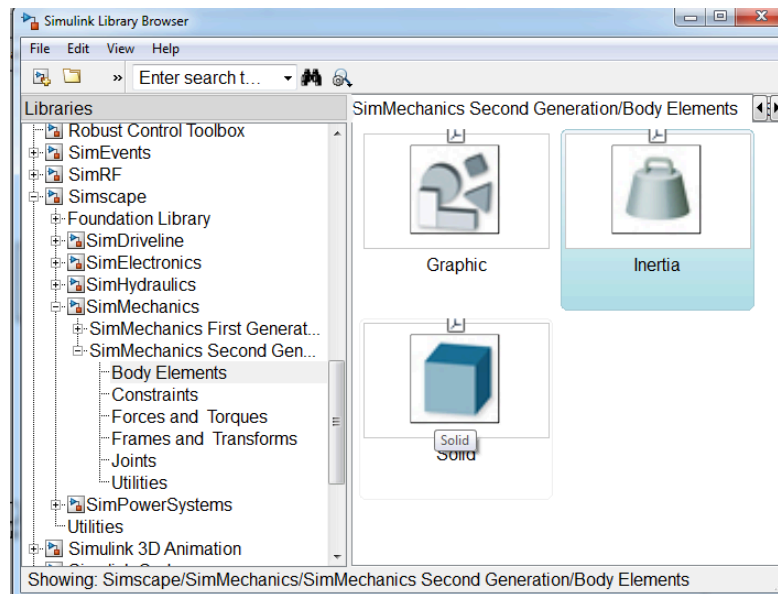
The model consists of solids, revolute joint, rigid transforms, world frame(reference), Simulink-PS converter and solver configuration. Consider the blocks in detail.

2.1. World frame



The block represents an inertial reference frame attached to the external environment. The frame is unique, and it provides an ultimate reference frame for a multibody system.

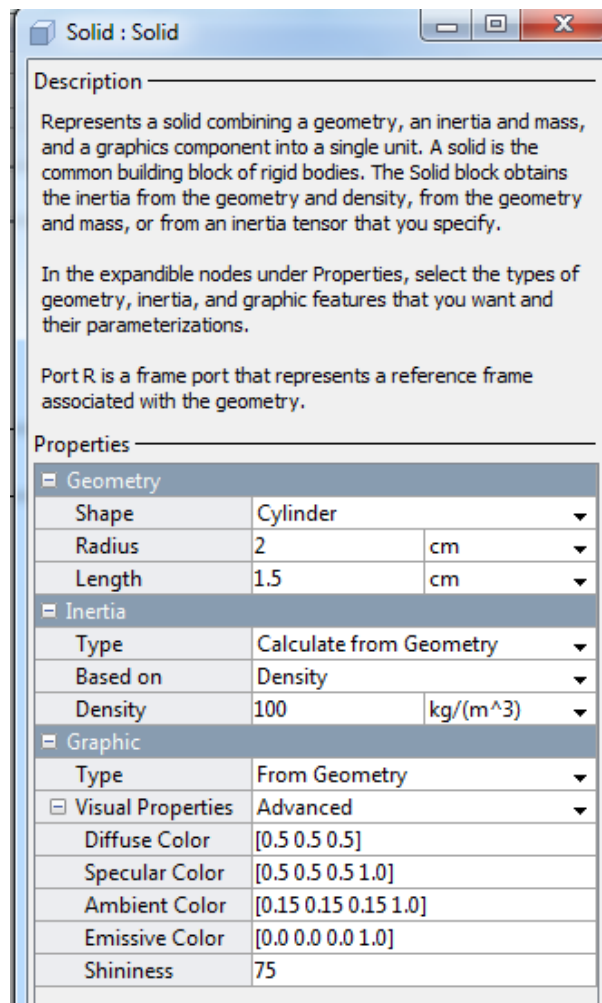
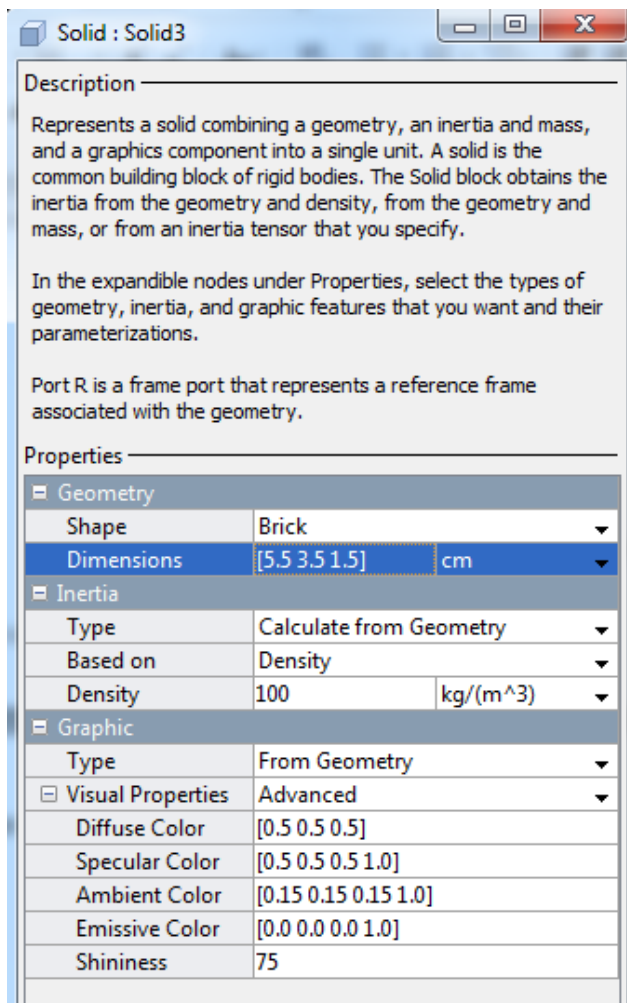
2.2. Solids



The block represents an elementary rigid mass. Block inputs include geometry, inertia, and graphic parameters. Use the block to represent a simple rigid body, or connect multiple Solid blocks to represent a compound rigid body. The Solid block assumes perfect rigidity with zero deformation under an applied force or torque.

Properties of

solids



Solid : Solid1

Description

Represents a solid combining a geometry, an inertia and mass, and a graphics component into a single unit. A solid is the common building block of rigid bodies. The Solid block obtains the inertia from the geometry and density, from the geometry and mass, or from an inertia tensor that you specify.

In the expandible nodes under Properties, select the types of geometry, inertia, and graphic features that you want and their parameterizations.

Port R is a frame port that represents a reference frame associated with the geometry.

Properties

Geometry

Shape

Brick

Dimensions

[2 1 7]

cm

Inertia

Type

Calculate from Geometry

Based on

Density

Density

100

kg/(m^3)

Graphic

Type

From Geometry

Visual Proper...

Simple

Color

[0.5 0.5 0.5]

Opacity

1.0

Solid : Solid2

Description

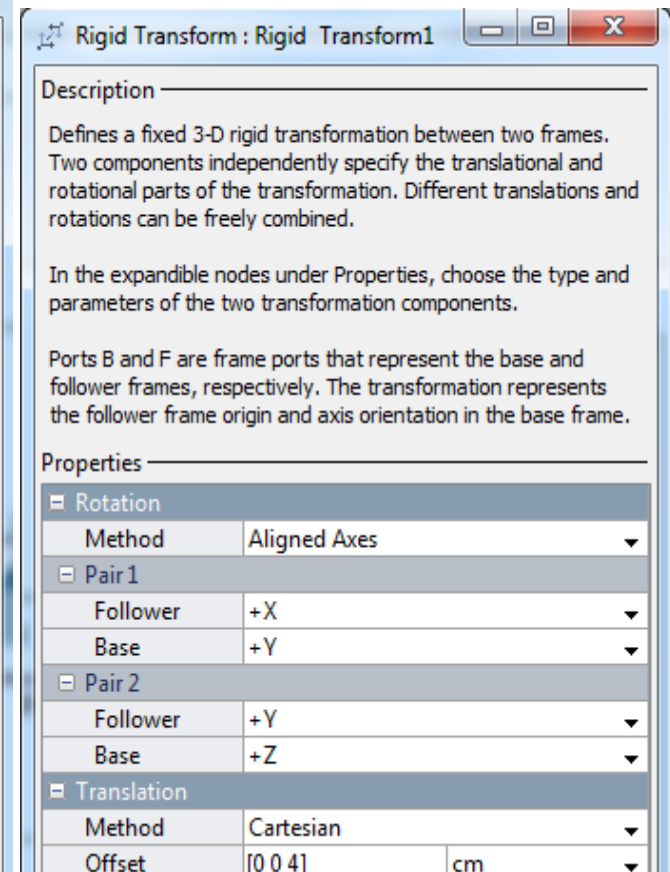
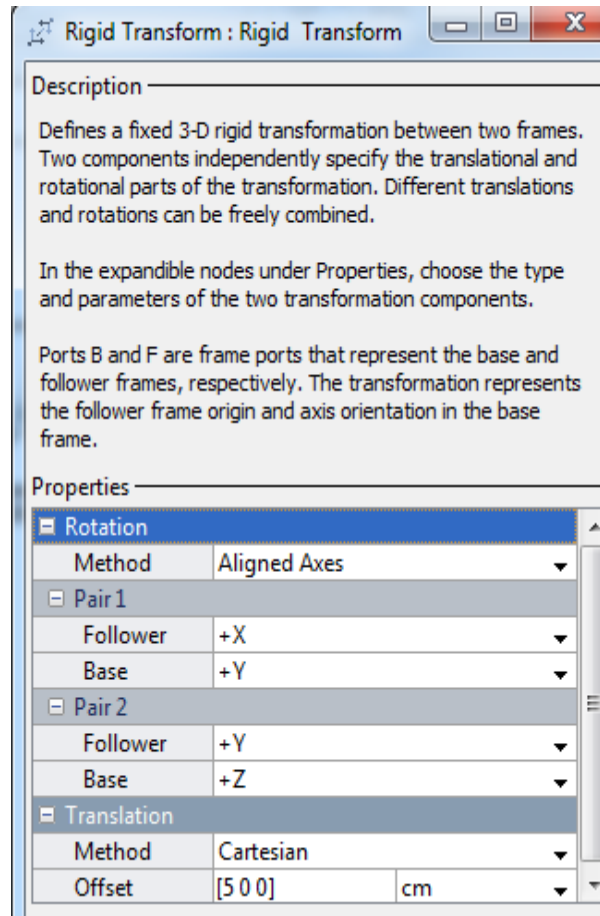
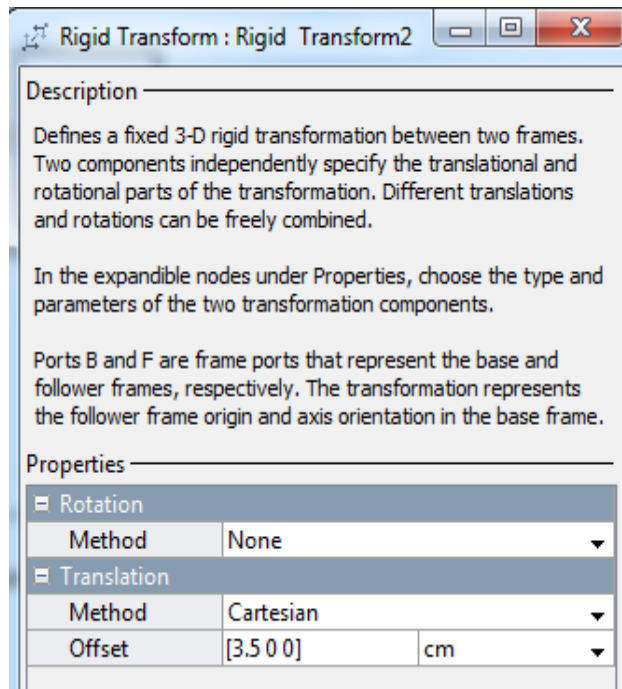
Represents a solid combining a geometry, an inertia and mass, and a graphics component into a single unit. A solid is the common building block of rigid bodies. The Solid block obtains the inertia from the geometry and density, from the geometry and mass, or from an inertia tensor that you specify.

In the expandible nodes under Properties, select the types of geometry, inertia, and graphic features that you want and their parameterizations.

Port R is a frame port that represents a reference frame associated with the geometry.

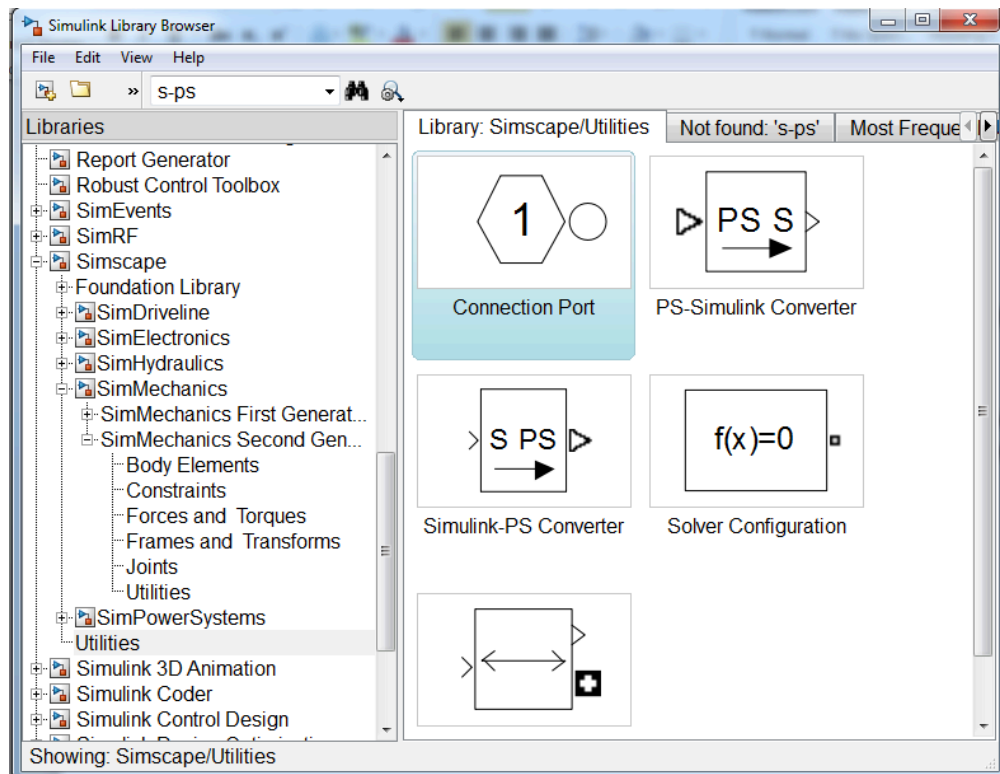
Properties

Geometry		
Shape	Sphere	
Radius	1.5	cm
Inertia		
Type	Calculate from Geometry	
Based on	Density	
Density	100	kg/(m^3)
Graphic		
Type	From Geometry	
Visual Properties	Simple	
Color	[0.5 0.5 0.5]	
Opacity	1.0	



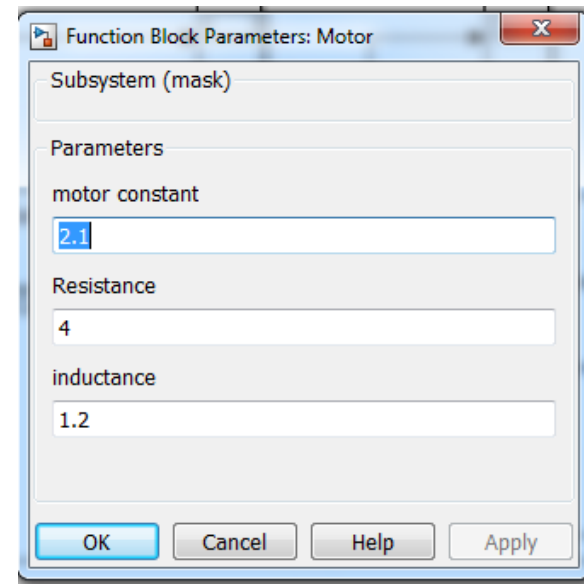
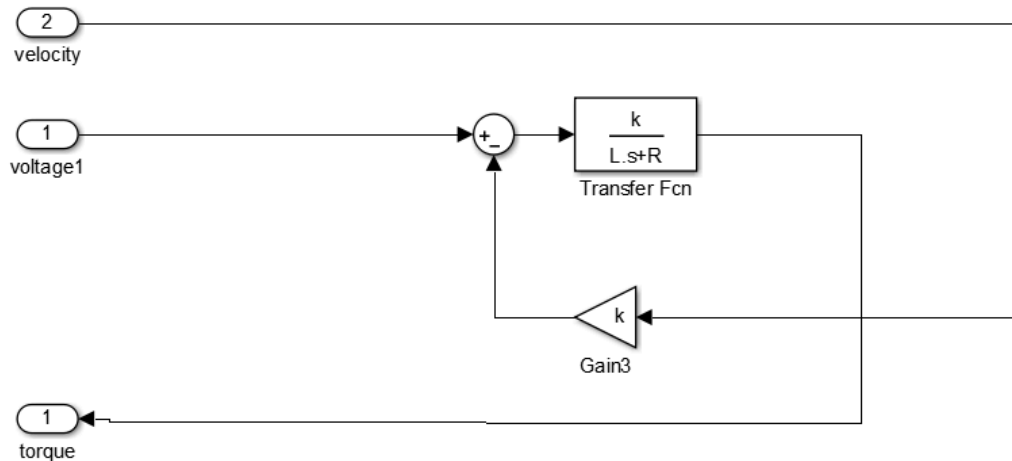
2.3. Simulink-PS converter

The Simulink-PS Converter block converts the input Simulink signal into a physical signal. Use this block to connect Simulink sources or other Simulink blocks to the inputs of a Physical Network diagram.



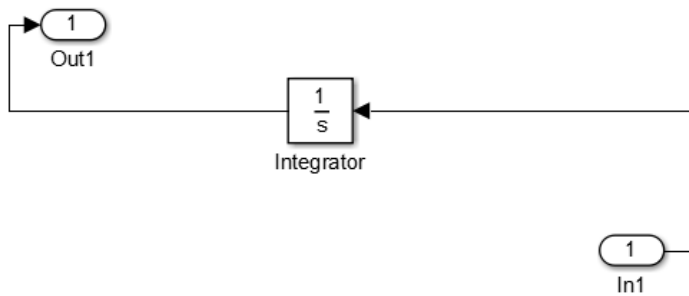
3. Motor

A model of the motor is used. Otherwise motor blocks in simpowersystems can be used.

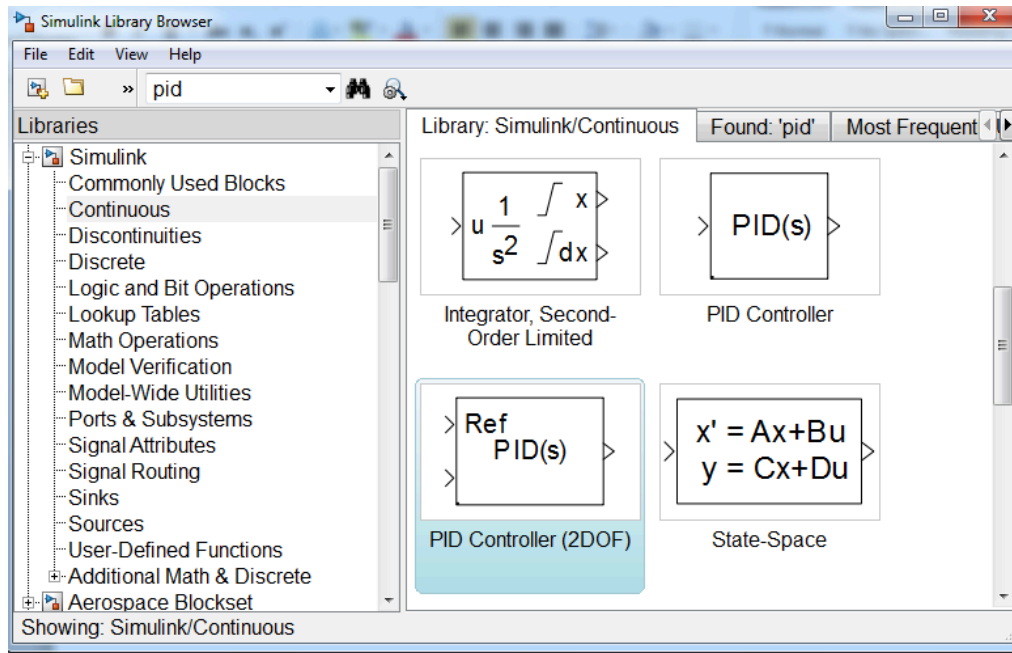


Note: study the exercise on how to create a mask

4. Encoder-integrate the velocity to obtain angle of rotation



5. PID controller



Implement a continuous- or discrete-time two-degree-of-freedom controller (PID, PI, or PD) in your Simulink model. The PID Controller (2DOF) block allows you to implement setpoint weighting in your controller to achieve both smooth setpoint tracking and good disturbance rejection. The PID Controller (2DOF) block generates an output signal based on the difference between a reference signal and a measured system output. The block computes a weighted difference signal for each of the proportional, integral, and derivative actions according to the setpoint weights you specify. The block output is the sum of the proportional, integral, and derivative actions on the respective difference signals, where each action is weighted according to the gain parameters. A first-order

pole filters the derivative action. Controller gains are tunable either manually or automatically.

Automatic tuning requires Simulink Control Design software (PID Tuner or SISO Design Tool). Configurable options in the

PID Controller (2DOF) block include:

- Controller type (PID, PI, or PD)

- Controller form (Parallel or Ideal)

- Time domain (continuous or discrete)

- Initial conditions and reset trigger

- Output saturation limits and built-in anti-windup mechanism

Function Block Parameters: PID Controller (2DOF)

PID Controller (2DOF)

This block implements continuous- and discrete-time PID control algorithms with setpoint weighting and includes advanced features such as anti-windup, external reset, and signal tracking. You can tune the PID gains automatically using the 'Tune...' button (requires Simulink Control Design).

Controller: PID Form: Parallel

Time domain:

☒ Continuous-time

☐ Discrete-time

Main PID Advanced Data Types State Attributes

Controller parameters

Proportional (P): 19.641641022791 ⊞ [Compensator formula](#)

Integral (I): 1.46683392409806

Derivative (D): -1.08856717563683

Filter coefficient (N): 18.0435727462574

Setpoint weight (b): 1

Setpoint weight (c): 1

Tune...

$$P(b \cdot r - y) + I \frac{1}{s} (r - y) + D \frac{N}{1 + N \frac{1}{s}} (c \cdot r - y)$$

Initial conditions

Source: internal

Integrator: 0

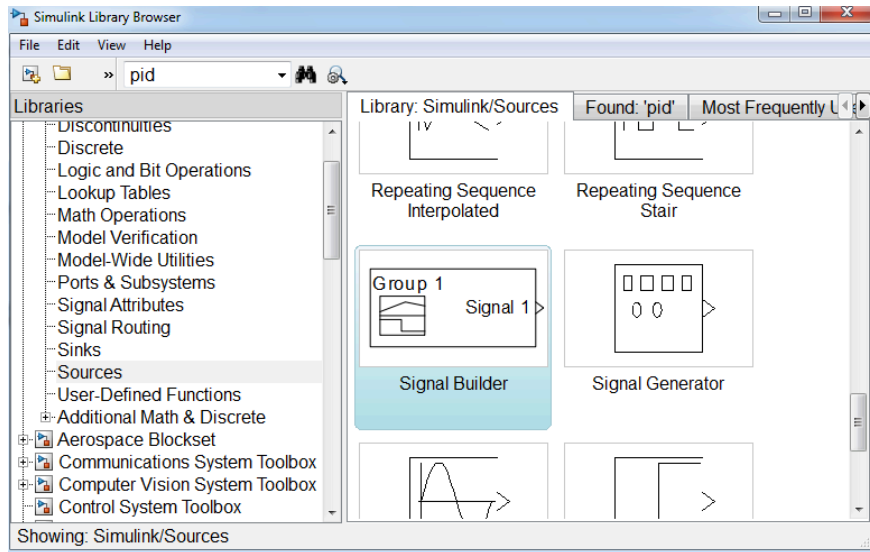
Filter: 0

External reset: none

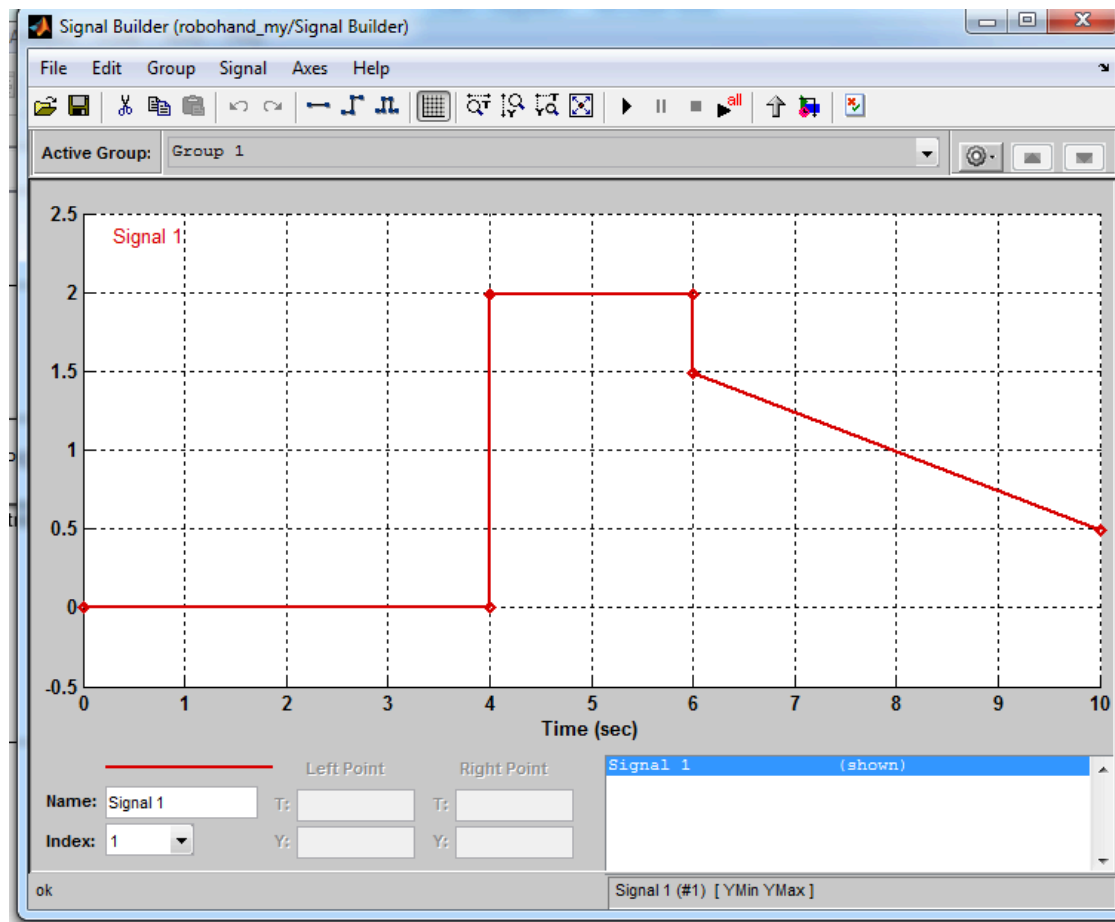
☐ Ignore reset when linearizing

☒ Enable zero-crossing detection

6. Reference signal



The Signal Builder block allows you to create interchangeable groups of piecewise linear signal sources and use them in a model.



Assignment: as given by instructor

Simulate for a 2 link 2 joint Robot
